Worksheet 9

due at 23:59 November 16th (to be discussed on November 17)

1 Controlled Modular Exponentiation

Consider the modular multiplication operator given by:

$$U_x |y\rangle = \begin{cases} |x\ y\ \ mod\ N\rangle & 0 \leq y < N \\ |y\rangle & N \leq y < 2^L \end{cases} \tag{1}$$

where x and N are coprime and $L = \lceil log_2 N \rceil$. Let the counting register consist of t qubits with computational basis states $|z\rangle = |z_{t-1}...z_1 z_0\rangle$, $z_j \in \{0, 1\}$. For each $j \in \{0, ..., t-1\}$, consider we apply the controlled unitary operation $C - U_x^{2^j}$ which has for control the j-th qubit in the counting register and the work register as the target and it applies $U_x^{2^j}$ if the control qubit is in the state $|1\rangle$.

(So if the counting register is in state $|0...000\rangle$, the identity operation is applied on the work register, if it's in the state $|0...001\rangle$ then U_x^1 is applied and so on).

1) Show that the successive application of these controlled unitaries implements:

$$|z\rangle_c |y\rangle_w \rightarrow \begin{cases} |z\rangle_c |x^z\ y\ mod\ N\rangle_w & 0 \leq y < N \\ |z\rangle_c |y\rangle_w & N \leq y < 2^L \end{cases} \tag{2}$$

The order finding algorithm's initial state $|\psi_0\rangle$ comprises of $|0\rangle = |0\rangle^{\otimes t}$ in the counting register and $|1\rangle$ in the work register comprising of $L = \lceil log_2 N \rceil$ qubits. Hadamard gates are applied to the qubits of the counting register.

2) Write down the state $|\psi_1\rangle$ after the Hadamard gates are applied.

The algorithm then applies the operation given in equation 2.

3) What is the state $|\psi_2\rangle$ after this step.

4) Show that you would obtain the same state if you apply a different unitary transform V but initialize the work register in the state $|0\rangle$, where V is given by:

$$V |j\rangle |k\rangle = \begin{cases} |j\rangle |k + x^j\ mod\ N\rangle & 0 \leq k < N \\ |j\rangle |k\rangle & N \leq k < 2^L \end{cases} \tag{3}$$

2 Finding the Order

The state $|\psi_2\rangle$ which you got in question (3) of exercise 1 may be written as:

$$|\psi_2\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left[\frac{1}{\sqrt{2^t}} \sum_{z=0}^{2^t-1} e^{2\pi i (s/r) z} |z\rangle_c \right] \otimes |u_s\rangle_w \tag{4}$$

where r is the order and $|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |x^k \ mod \ N\rangle$ is an eigenstate of the unitary in equation 1.

1) Derive this expression (hint: use the fact that the state $|1\rangle$ may be written as a sum over these eigenstates, theorem 8.5 in the lecture notes).

2) The lecture notes mention that the step which uses theorem 8.5 to arrive at this expression is "one of the prettiest in the course" (page 37). Explain why writing $|\psi_2\rangle$ in this fashion is crucial to the order finding problem.

At the end of the order finding algorithm (after applying an inverse QFT on the register), we measure the counting register and we get an integer m which is a t-bit approximation of $2^t s/r$ for a uniformly random $s \in \{0, ..., r-1\}$. Finally, to find the order, we use the continued fractions algorithm.

3) Using theorems 9.3 and 5.2, determine the number of qubits required in the counting register assuming a fixed failure probability ϵ.

4) The denominator we obtain from the continued fractions algorithm returns r' which only corresponds to the true r under some condition. What is that condition?

5) Suppose $r = 6$, list the possible values of s/r and identify which measurements allow us to obtain r. Assuming we have sufficiently accurate phase estimation, what is the probability that we would need to run the circuit exactly three times to obtain r?

3 From Order Finding to Factoring

Let N be a composite integer and x an integer coprime to N. Assume r is the order of $x \ mod \ N$.

1) Suppose r is even. Explain how knowing r allows us to find non-trivial factors of N. Under what condition does this procedure succeed?

2) Suppose we ran Shor's algorithm for a specific x which we picked uniformly at random from $\{2, ..., N-1\}$ and $gcd(x, N) = 1$. However, the r that we got is odd. Instead of choosing a new integer at random, can we pick x^k where $k > 1$ for which we get an even order? (hint: $ord_N(x^k) = \frac{r}{gcd(r,k)}$ when $gcd(x, N) = 1$ where $r = ord_N(x)$).

3) Consider $N = 15$ and $x = 14$. Determine the order r of $x \ mod \ N$. Show

that, although r is even, we do not get non-trivial factors of 15. Explain why this is the case.

4 Factoring 91

Algorithm 7.8 in the lecture notes presents Shor's factoring algorithm. What we have discussed in the exercises 1 and 2 were about step 5 of this algorithm which is the only quantum part. Let us now use Shor's algorithm to factor $N = 91$.

1) Verify that the algorithm reaches step 4 for an input $N = 91$.

Consider we are at step 4 now and that from $\{2, ..., N-1\}$, we picked uniformly at random $x = 4$ which happens to be coprime to 91. Thus we can move to step 5.

2) Consider we ran the order finding circuit and found $r = 6$. Verify that this is the order of 4 *mod* 91. Explain why the conditions for obtaining nontrivial factors are satisfied. Compute the relevant greatest common divisors using Euclid's algorithm and determine the factors of 91.

3) This question is independent from previous questions: Show that $N = 15$ is the smallest number for which the order finding subroutine is reached.

5 Complexity of Shor's Algorithm

Recall the modular multiplication operator defined in equation 1. In the order finding circuit, we apply the controlled operation $C - U_x^{2^j}$ for $j = 0, ..., t-1$ where t is the number of qubits in the counting register.

1) Assuming we implement each controlled operation using 2^j controlled applications of U_x. In terms of t, how many controlled applications of U_x do we need?

2) Explain how we improve this naive approach (hint: $U_x^{2^j} = U_{a_j}$ where $a_j = x^{2^j} \ mod \ N$). How does the improve this computational complexity of modular exponentiation?